

Fundamentals of Linear Control:

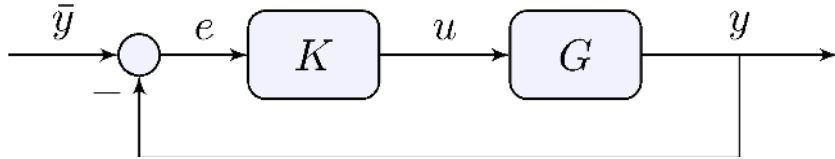
A Concise Approach

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Chapter 8: Performance and Robustness

linearcontrol.info/fundamentals

8.1 Closed-Loop Stability and Performance



Tracking performance

$$e = S\bar{y}, \quad S = \frac{1}{1 + L}, \quad L = GK$$

- Better performance needs $|S(j\omega)|$ small
- Because $H(s) + S(s) = 1$

$$S(j\omega) \approx 0 \text{ means } H(j\omega) \approx 1$$

Beware of measurement noise

- Cannot make $S(j\omega)$ small for all ω

Tracking performance

$$e = S\bar{y}, \quad S = \frac{1}{1 + L}, \quad L = GK$$

High gains makes S small

$$|L(j\omega)| > M \gg 1 \quad \implies \quad |S(j\omega)| < \frac{1}{M - 1}$$

Small gains make S one

$$|L(j\omega)| \approx 0 \quad \implies \quad |S(j\omega)| \approx 1$$

e.g. L is strictly proper

$$\lim_{|s| \rightarrow \infty} |L(s)| = 0 \quad \implies \quad \lim_{|s| \rightarrow \infty} |S(s)| = 1$$

Tracking performance

$$e = S\bar{y}, \quad S = \frac{1}{1 + L}, \quad L = GK$$

What if $|L(j\omega)| \approx 1$?

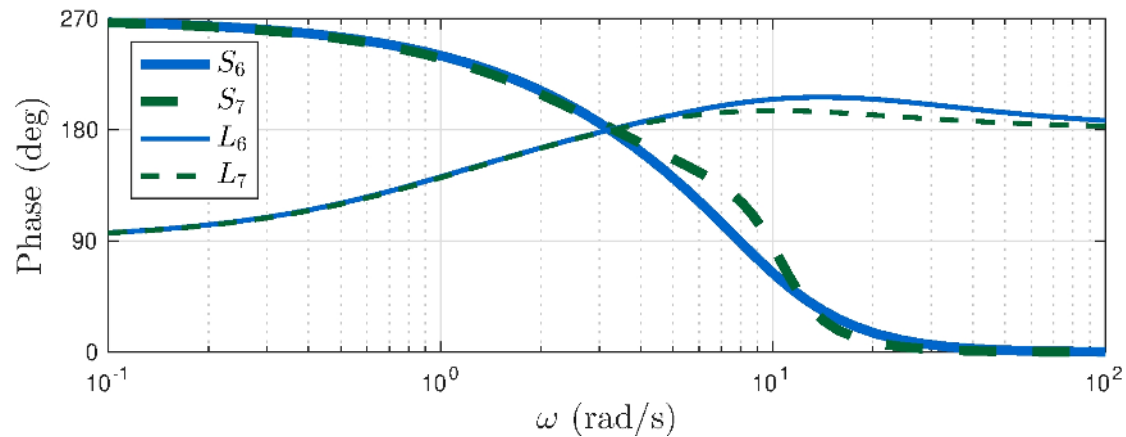
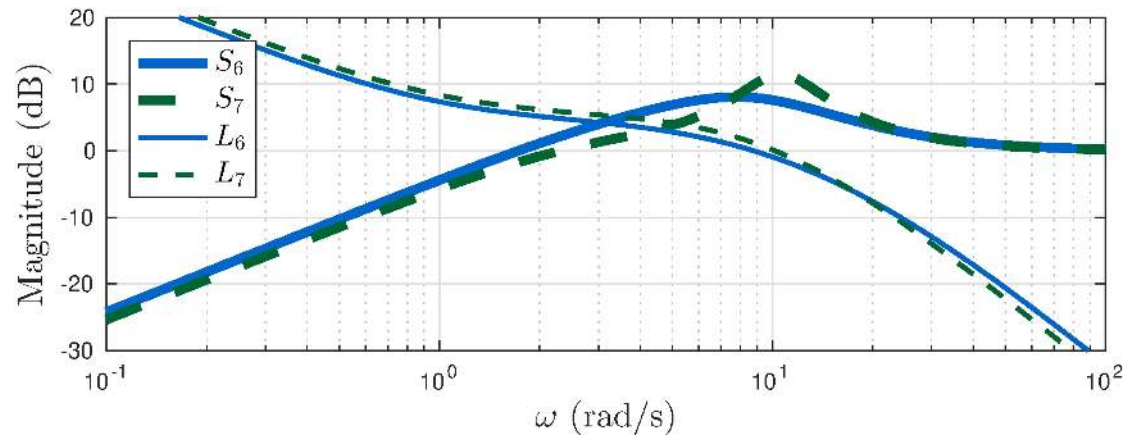
Look at the phase:

$$|L(j\omega)| \approx 1, \quad \angle L(j\omega) \approx \pi \quad \implies \quad |S(j\omega)| \text{ is large}$$

S large makes S_M small

$$S_M = 1 / \sup_{\omega} |S(j\omega)|$$

Example: control of the simple pendulum



What happens when $|L(j\omega)| \approx 1$?

Bode Sensitivity Integral

Theorem 8.1:

L has poles $p_i, \operatorname{Re}(p_i) > 0, i = 1, \dots, k$. If

$$\lim_{|s| \rightarrow \infty} s L(s) = \kappa, \quad |\kappa| < \infty$$

and $S = (1 + L)^{-1}$ is asymptotically stable then

$$\int_0^\infty \ln |S(j\omega)| d\omega = \int_0^\infty \ln \frac{1}{|1 + L(j\omega)|} d\omega = \pi \sum_{i=1}^k p_i - \frac{\pi \kappa}{2}$$

Main takeaways:

- Closed-loop sensitivity is on a balanced budget
- Unstable systems are hard
- Choose your battles wisely!

See book for more:

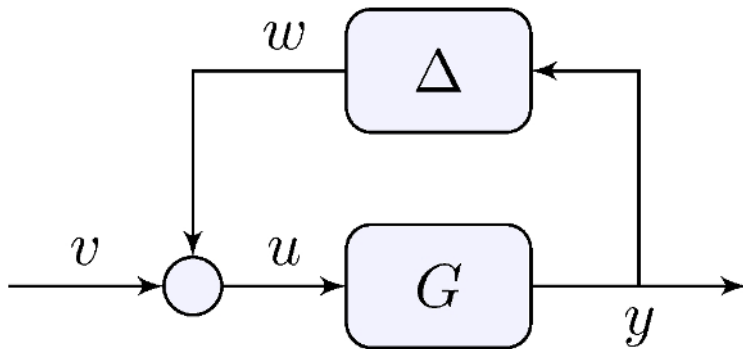
- How limited bandwidth makes things harder
- How unstable zeros amplify the effect of unstable poles

8.2 Robustness

Margins

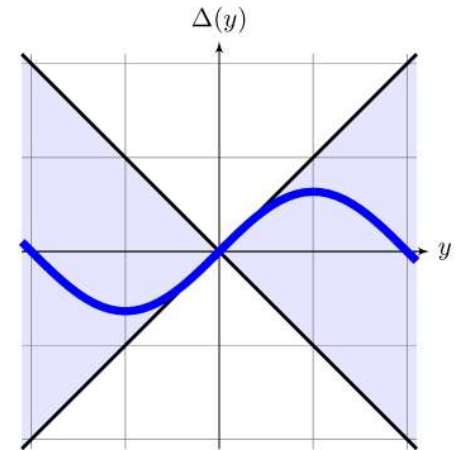
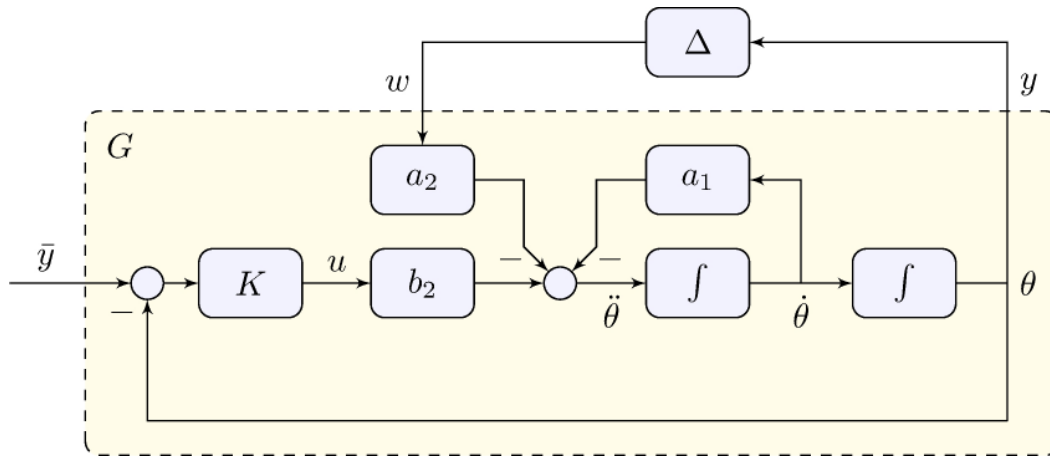
- Gain margin: robustness to changes in gain
- Phase margin: robustness to changes in phase

Robustness analysis



Robustness = asymptotic stability for all $\Delta \in \Delta$

Example: control of the simple pendulum



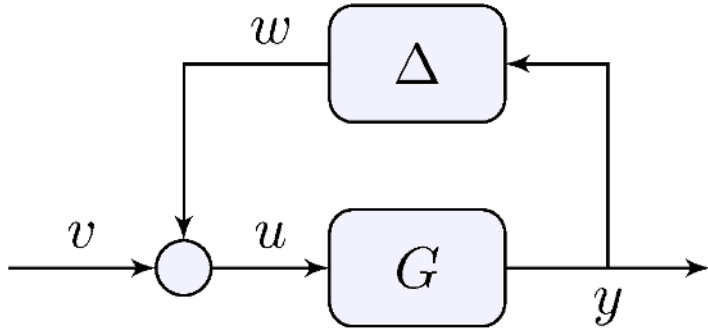
Nonlinearity as an “uncertainty”

$$\Delta(y) = \sin(y)$$

Uncertainty model

$$\Delta = \{\Delta : \mathbb{R} \rightarrow \mathbb{R}, \quad |\Delta(y)| \leq |y|\}$$

8.3 Small Gain



$$\Delta = \{\Delta : \mathbb{R} \rightarrow \mathbb{R}, \quad |\Delta(y)| \leq |y|\}$$

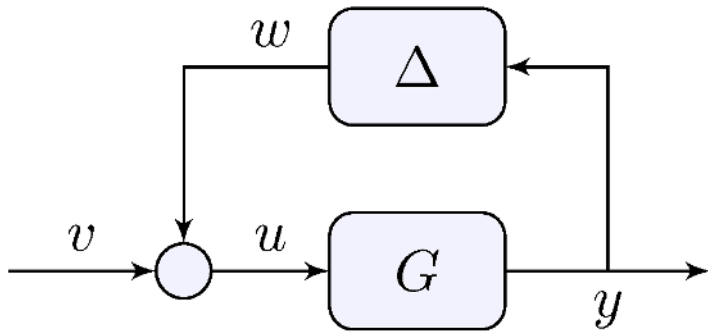
Disturbance

$$\|w\|_2 = \|\Delta(y)\|_2 \leq \|y\|_2 \quad \text{for all } \Delta \in \Delta$$

System

$$\|y\|_2 \leq \|G\|_\infty (\|w\|_2 + \|v\|_2) + M\|x_0\|_2, \quad \|G\|_\infty = \sup_{\omega \in \mathbb{R}} |G(j\omega)|$$

How about the closed-loop?



$$\Delta = \{\Delta : \mathbb{R} \rightarrow \mathbb{R}, \quad |\Delta(y)| \leq |y|\}$$

Small gain theorem

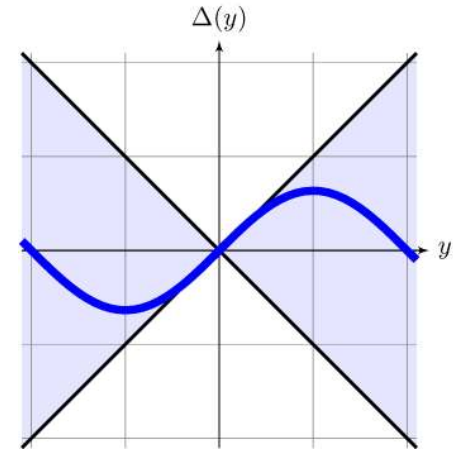
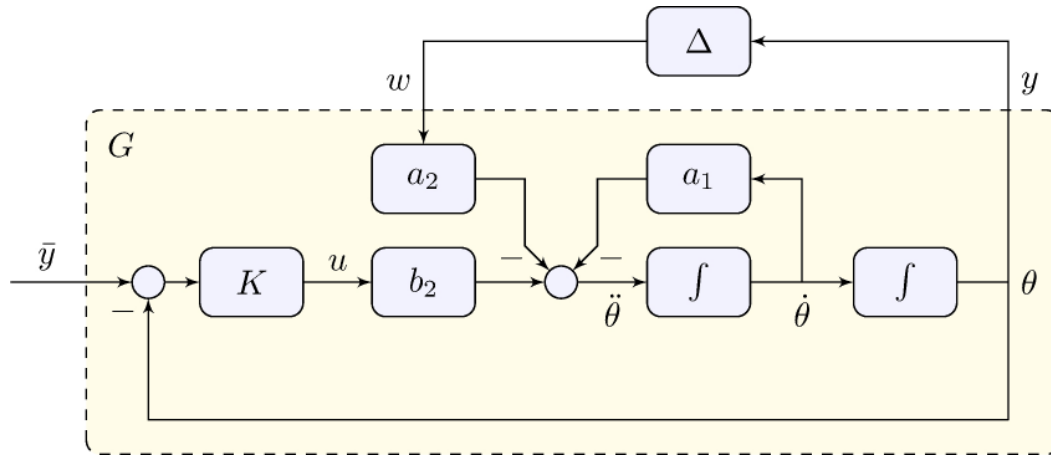
If G is asymptotically stable and $\|G\|_\infty < 1$ then

$$\|y\|_2 \leq \frac{\|G\|_\infty}{1 - \|G\|_\infty} \|v\|_2 + \frac{M}{1 - \|G\|_\infty} \|x_0\|_2$$

See book for more:

- Dynamic uncertainty
- Delays

8.4 Control of the Simple Pendulum - Part III



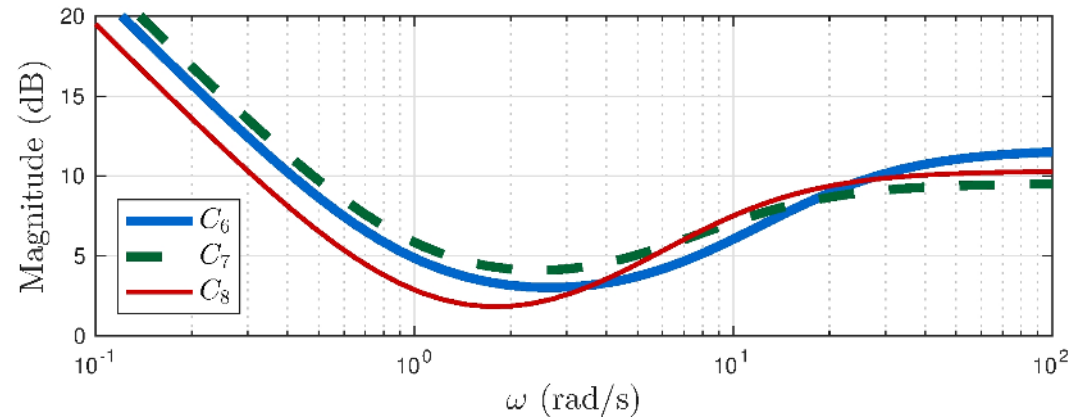
Model for robustness analysis

$$G(s) = \frac{Y(s)}{W(s)} = \frac{-a_2 F(s)}{1 + b_2 K(s) F(s)}, \quad F(s) = \frac{1}{s^2 + a_1 s}$$

Data

$$a_1 = \frac{b}{J_r} = 0, \quad a_2 = \frac{m g r}{J_r} \approx 49 \text{s}^{-2}, \quad b_2 = \frac{1}{J_r} \approx 66.7 \text{kg}^{-1} \text{m}^{-2}$$

Small gain analysis ($\bar{y} = 0$)

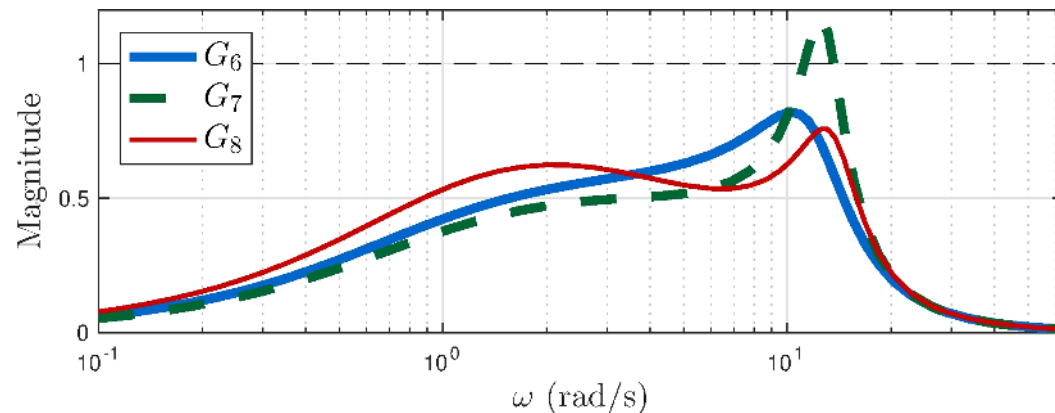


$$C_6(s) = 3.83 \frac{(s + 0.96)(s + 6.86)}{s(s + 21)}$$

$$C_7(s) = 3 \frac{(s + 1)(s + 5)}{s(s + 11)}$$

$$C_8(s) = 3.28 \frac{(s + 1)(s + 3)}{s(s + 10.5)}$$

- C_8 : higher gains at higher frequency to lower $\|G_8\|_\infty$

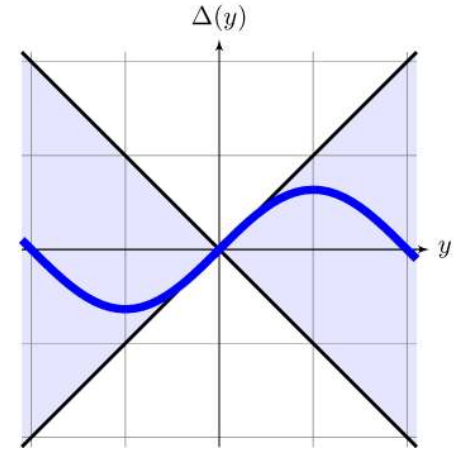
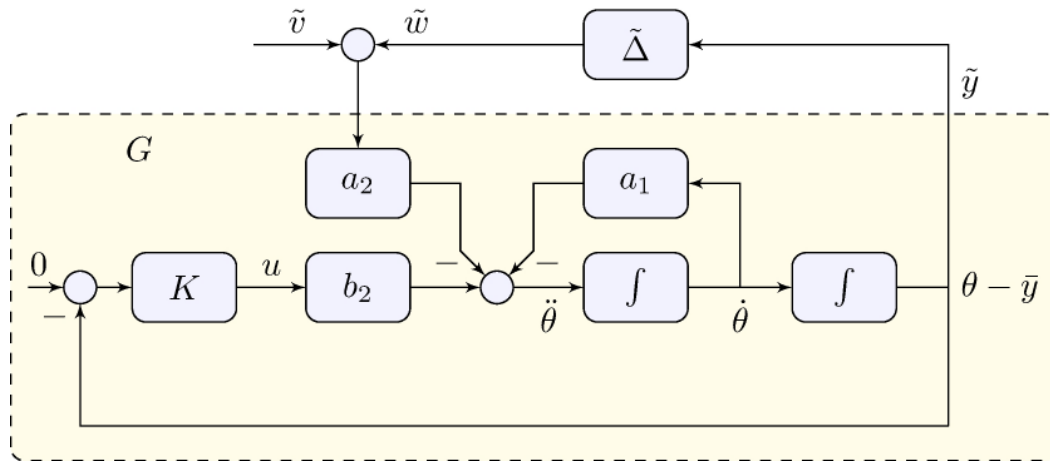


$$\|G_6\|_\infty \approx 0.82$$

$$\|G_7\|_\infty \approx 1.17$$

$$\|G_8\|_\infty \approx 0.76$$

Small gain analysis ($\bar{y} \neq 0$)



Nonlinearity

$$\tilde{\Delta}(\tilde{y}) = \cos(\bar{y}) \sin(\tilde{y}) + \sin(\bar{y})(\cos(\tilde{y}) - 1), \quad \tilde{v} = \sin(\bar{y})$$

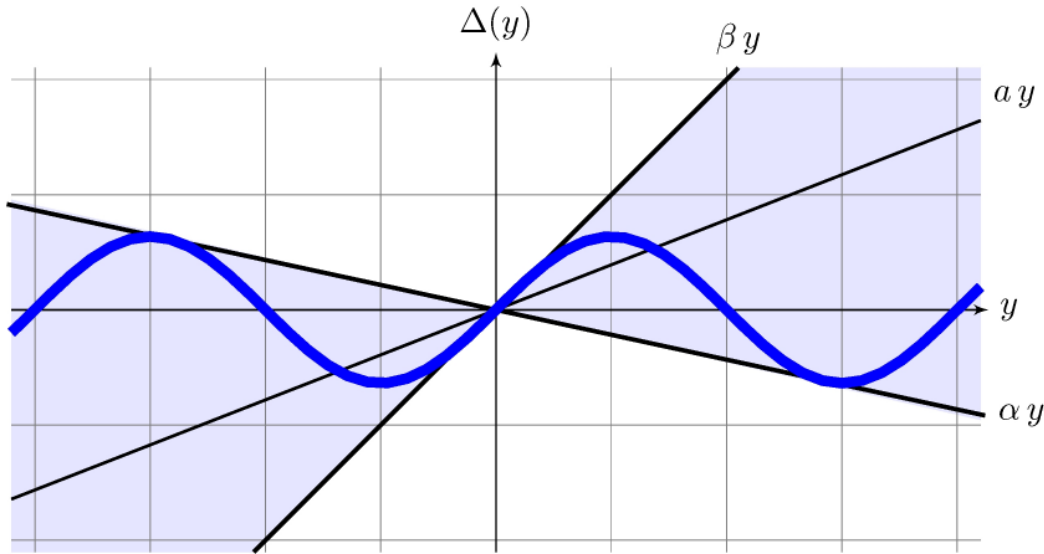
Uncertainty model

$$\tilde{\Delta} = \Delta = \{\Delta : \mathbb{R} \rightarrow \mathbb{R}, \quad |\Delta(y)| \leq |y|\}$$

Stability analysis holds also for tracking!

See book for details and more

8.5 Circle Criterion



Sector nonlinearities

$$\Delta_{ab}(a, b) = \{\Delta : \mathbb{R} \rightarrow \mathbb{R}, \quad |\Delta(y) + ay| \leq b|y|, \quad b > 0\},$$

$$(a, b) = (\tfrac{1}{2}(\alpha + \beta), \tfrac{1}{2}(\beta - \alpha))$$

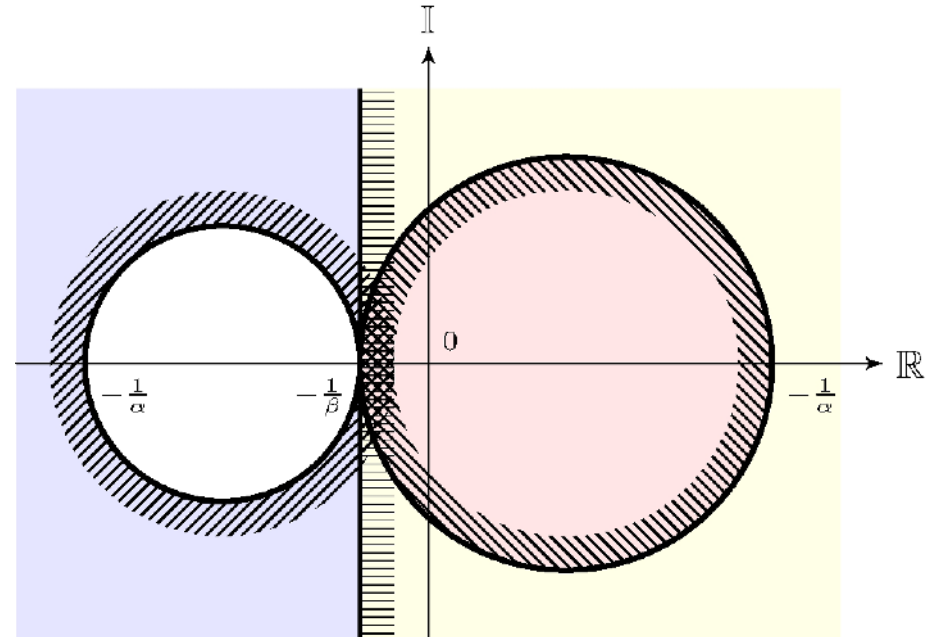
$$\Delta_{\alpha\beta}(\alpha, \beta) = \{\Delta : \mathbb{R} \rightarrow \mathbb{R}, \quad (\Delta(y) - \alpha y)(\Delta(y) - \beta y) \leq 0, \quad \beta > \alpha\},$$

$$(\alpha, \beta) = (a - b, a + b)$$

Circle Criterion

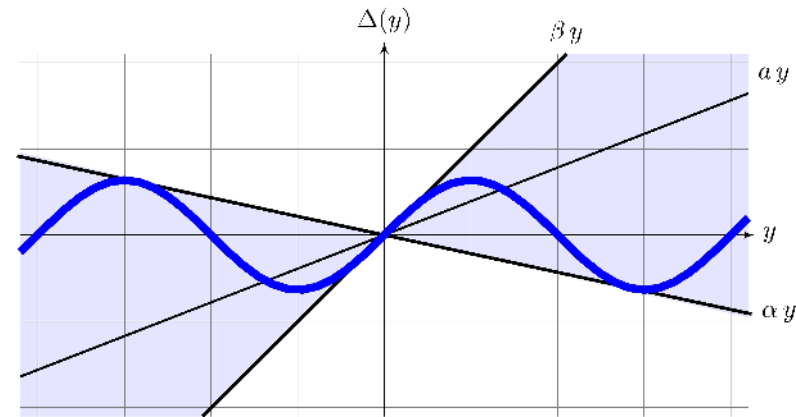
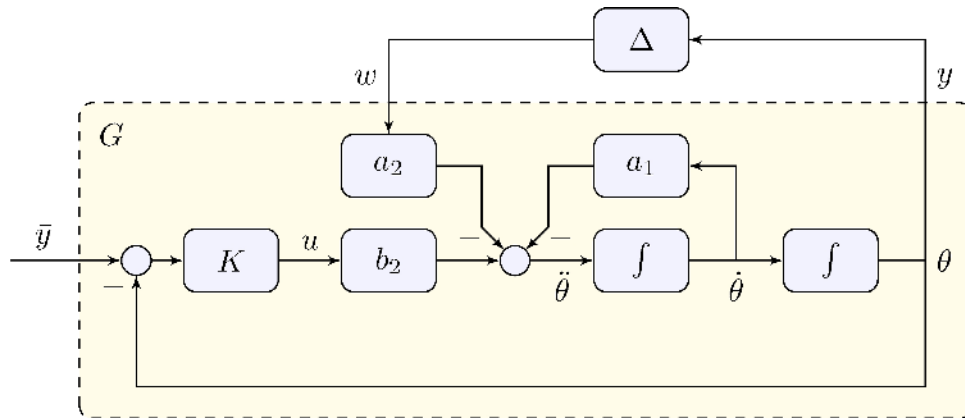
Theorem 8.2

Origin is globally asymptotically stable if either



1. $\beta > 0 > \alpha$, G is asymptotically stable and the polar plot of G is inside $C(\alpha, \beta)$;
 2. $\beta > \alpha > 0$, the Nyquist plot of G encircles $C(\alpha, \beta)$ m times but never enters $C(\alpha, \beta)$.
- m is the number of unstable poles of G

Example: control of the simple pendulum



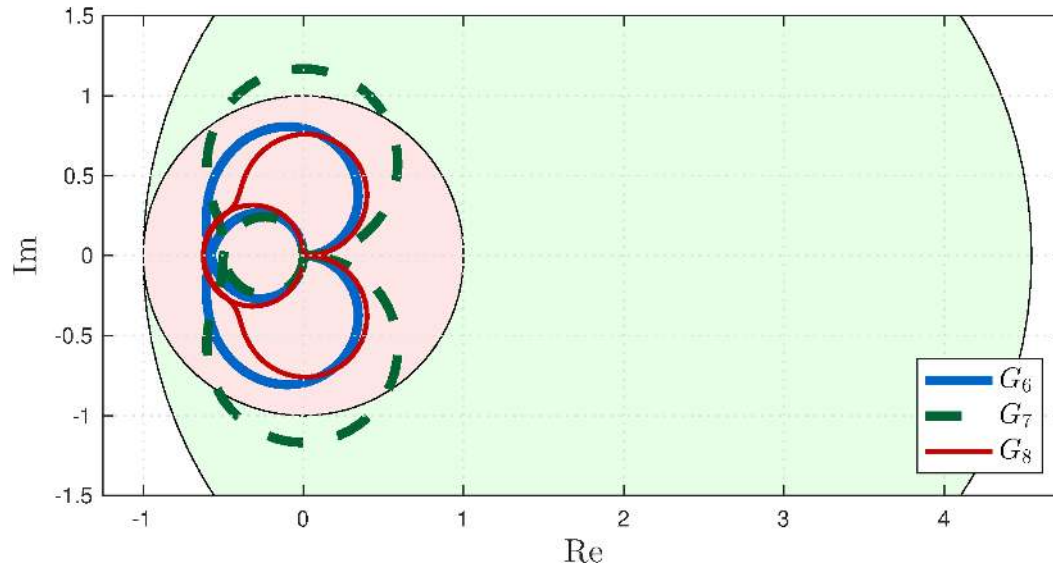
Model for robustness analysis

$$G(s) = \frac{Y(s)}{W(s)} = \frac{-a_2 F(s)}{1 + b_2 K(s) F(s)}, \quad F(s) = \frac{1}{s^2 + a_1 s}, \quad \alpha = -0.22, \quad \beta = 1$$

Data

$$a_1 = \frac{b}{J_r} = 0, \quad a_2 = \frac{m g r}{J_r} \approx 49 \text{s}^{-2}, \quad b_2 = \frac{1}{J_r} \approx 66.7 \text{kg}^{-1} \text{m}^{-2}$$

Example: control of the simple pendulum



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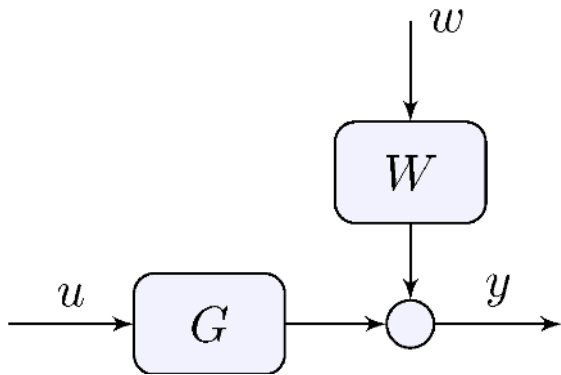
Small gain

- $C(-1, 1)$
- G_7 fails

Circle criterion

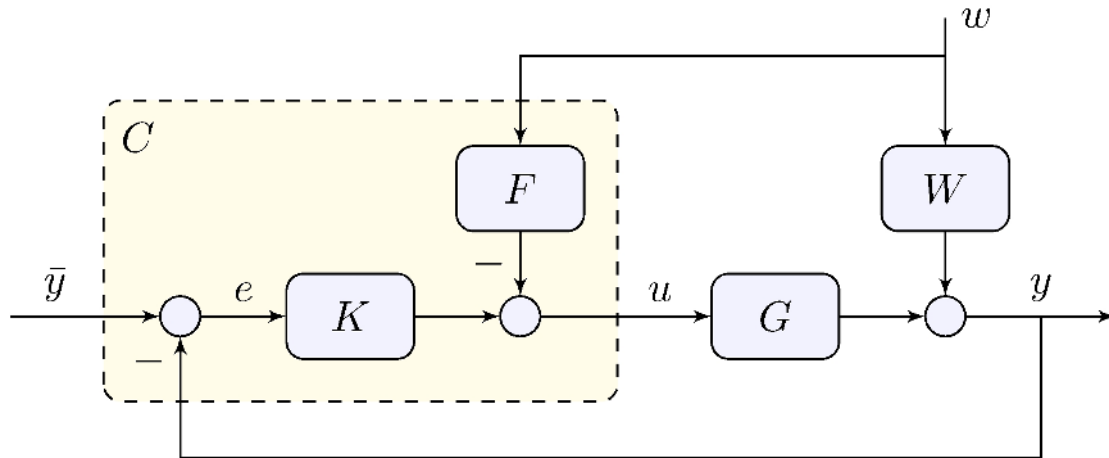
- $C(-0.22, 1)$
- All controllers stabilizing

8.6 Feedforward Control and Filtering



If the “disturbance” w is known or can be measured *feedforward* can be combined with feedback

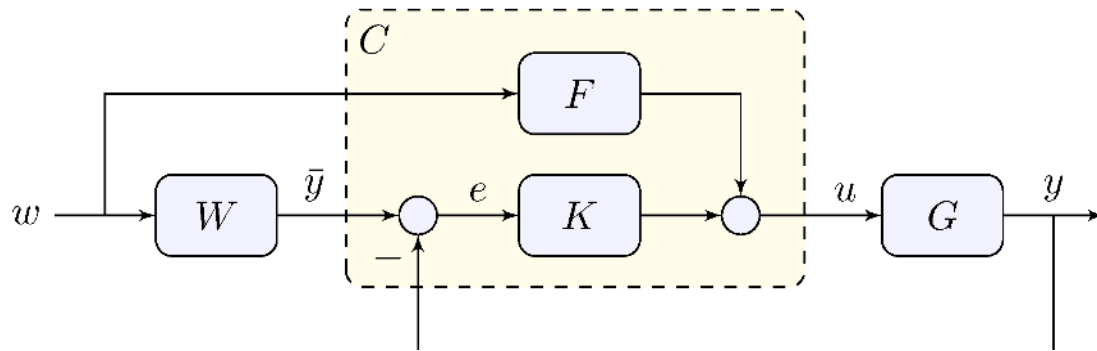
Feedforward control



Tracking error

$$e = S \bar{y} - (W - F G) S w, \quad S = (1 + G K)^{-1}$$

Reference feedforward control



Tracking error

$$e = (W - F G) S w, \quad S = (1 + G K)^{-1}$$

Feedforward control design

Beware: since F is in open-loop it has to be asymptotically stable!

If possible

$$F = G^{-1}W \quad \implies \quad W = FG$$

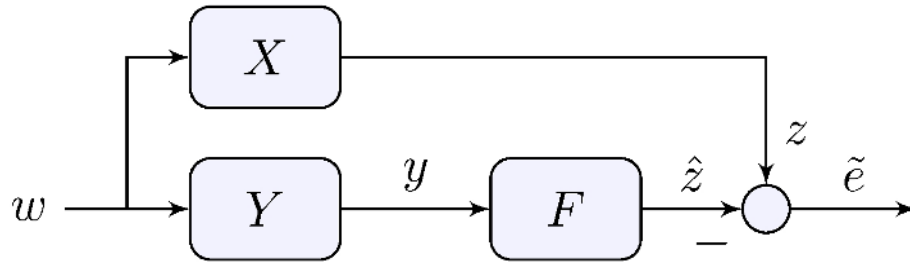
At a given frequency

$$F(j\omega) = \lim_{s \rightarrow j\omega} G(s)^{-1}W(s)$$

Optimal filter

$$\min_F \{ \| (W - FG)S \| : F \text{ is asymptotically stable and realizable} \}$$

Optimal filtering

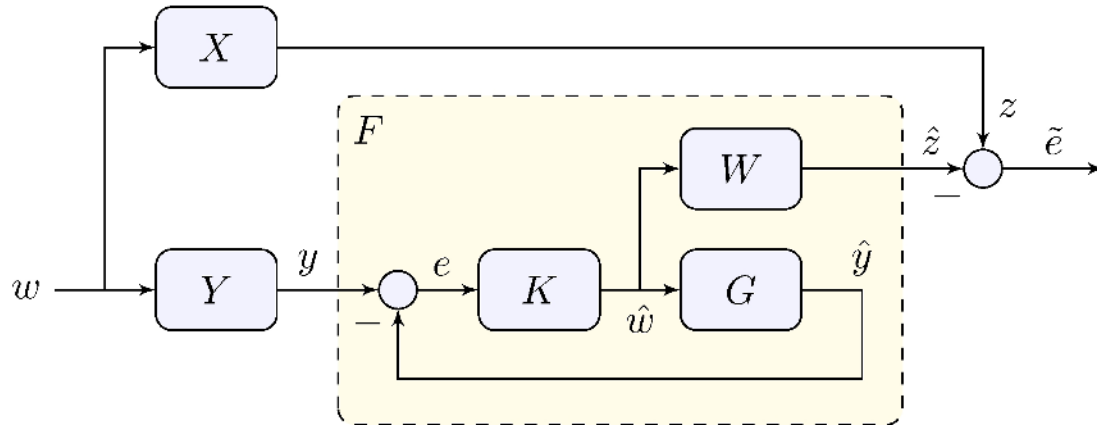


Optimization problem

$$\min_F \{ \|X - F Y\| : F \text{ is asymptotically stable and realizable} \}$$

See book for simple examples

Optimal filtering



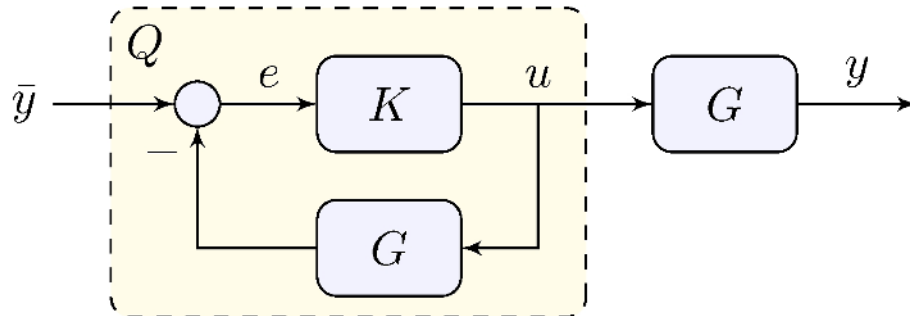
Solution via feedback

$$W = X, \quad G = Y, \quad \tilde{e} = W S w, \quad S = \frac{1}{1 + G K}$$

e.g. K or G have a pole at zero then $\lim_{t \rightarrow \infty} \tilde{e}(t) = 0$

This is similar to a *Kalman filter*

All Stabilizing Controllers



Assumption: G is asymptotically stable

$$K \text{ is stabilizing} \quad \implies \quad Q = \frac{K}{1 + K G} \text{ is asymptotically stable}$$

$$Q \text{ is asymptotically stable} \quad \implies \quad K = \frac{Q}{1 - Q G} \text{ is stabilizing}$$

A non-trivial version exists in which G does not need to be asymptotically stable

Learn more

- [Book website](#)

